Tech4Food 2026

Organised by Dept. of Computer Engineering, PCCOE, Pune · CESA · ACM PCCOE Student Chapter

ROUND 1 · IDEA SUBMISSION

Final Round	Venue	Host Institute
24 October 2026	PCCOE, Pune	Pimpri Chinchwad College of Engineering, Pune

Strictly Maintain no of slides under 9

SaafScan: 10-second adulterant screening on any phone camera

Iodine and methylene-blue strips, read by any phone camera

Team Members

1. [Name] (Team Lead)
2. [Name]
3. [Name]

Team Details	
Team Name	[ ]
Team Lead	[ ]
College Name	[ ]
Theme	DETECT · Food Safety Intelligence
Solution Type	Both: app + test strip

Problem Understanding

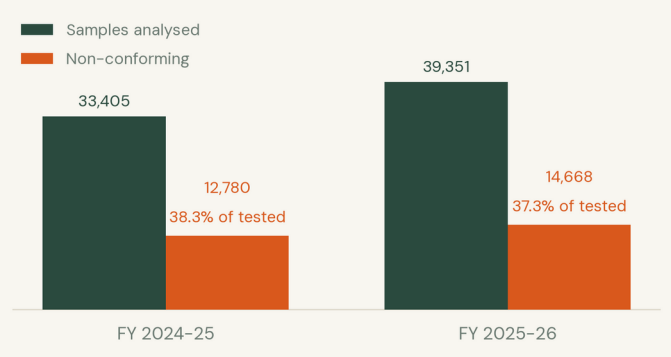
Loose milk is diluted with water, then thickened with starch and whitened with detergent so the dilution does not show. A buyer cannot check any of that before pouring it for a child.

866 M
foodborne illnesses a year worldwide

1.52 M
deaths a year linked to unsafe food

37.3%
of India's milk and dairy samples failed FSSAI tests in FY 2025-26

FSSAI SURVEILLANCE, TWO YEARS RUNNING



FSSAI data tabled in Parliament, Lok Sabha Unstarred Q 1092, 24 July 2026

What a buyer has today

A lab sample takes days and costs more than the milk. Rapid kit strips exist, but they read by eye, so two people get two answers and nobody keeps a record.

WHO THIS PUTS A CHECK IN THE HANDS OF

- Street vendors
- Canteens & hostels
- Dairy collection centres
- Mothers buying loose milk
- Mess committees
- Food inspectors

Proposed Solution & Innovation

A printed card with two reagent zones. One drop of the sample, one photo, one verdict with a confidence you can argue with.

01
Print the card
White, grey and black patches plus two reaction zones

02
Add one drop
Iodine for starch, methylene blue for detergent

03
Photograph it
Any phone camera, any light, no app store

04
Read the colour
The app compares each zone against the printed patches

05
Keep the record
Every test is logged and gets a PDF with a QR code

WHAT IT CALLS OUT

Starch | Detergent | Both | None | Inconclusive

Starch turns iodine blue-black. Detergent shifts methylene blue. Both zones are measured as ΔE against the printed reference patches.

The refusal to guess

Confidence is temperature-calibrated, so a reported 90% means 90%. Anything below the threshold returns no verdict at all and is marked inconclusive. A wrong answer a family acts on is worse than no answer.

AGAINST WHAT EXISTS

	Time to answer	Needs a lab	Keeps a record	Works offline
Lab test	days	yes	yes	no
Rapid kit strip	minutes	no	no	yes
SaafScan	under 10 s	no	yes, with QR	yes

Lab and rapid-strip columns are how those methods are described in the FSSAI and WHO material cited on slide 7; SaafScan timings are measured.

Technical Architecture

Camera or upload → Card calibration white/grey/black → Strip region auto-detected → Zone colour Lab + ΔE vs card → 12 colour and ΔE features → **Gradient boosting** → Calibrated confidence

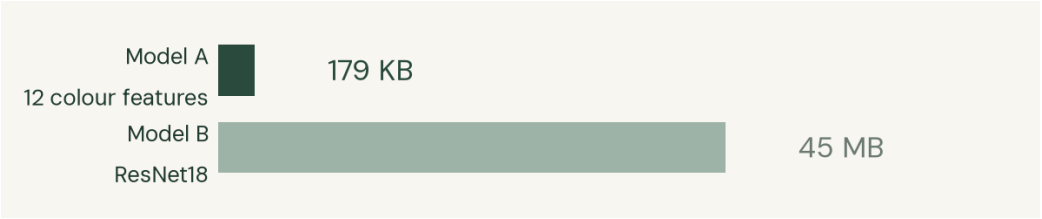
confidence ≥ 0.55 → verdict, severity and advice, or inconclusive and sent to a lab

Below threshold the pipeline writes an inconclusive entry, not a guess. A deterministic rule table stays reachable as a fallback if the ML runtime fails to load.

SERVES AND STORES

Append-only SQLite → PDF + QR report → Batch id on every test → Recall scoped to a batch → PWA frontend, no install

WHY THE SMALLER MODEL SHIPS

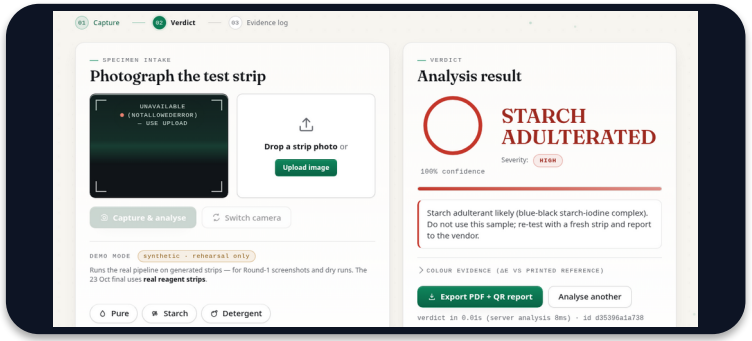


The model does not beat the rule table, so the deck does not claim it does

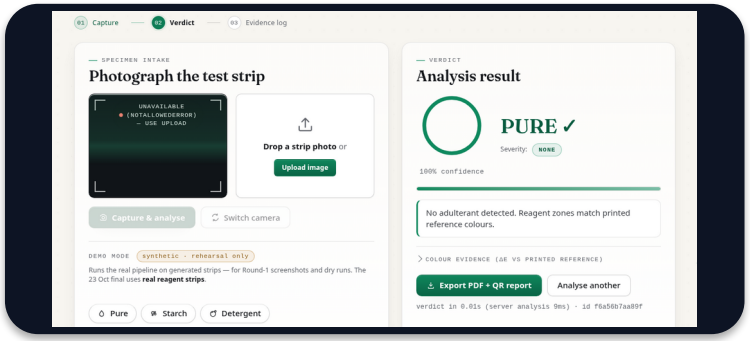
A 12-feature gradient booster and a ResNet18 were trained on the same split. On held-out lighting the booster scores 0.933 accuracy at 0.733 coverage, while the rule table scores 1.000 – a delta of -0.067, and the model abstains rather than guess. The ResNet18 abstains on all 45 held-out samples at its 0.999 threshold. The booster ships anyway: 179 KB against 45 MB, and it answers at 0.55 confidence, which leaves room to still answer when a real phone camera shifts the colour.

Stack: Python 3.12 · OpenCV · scikit-learn · ONNX Runtime on CPU · FastAPI · SQLite · browser frontend. No GPU is needed at inference.

Implementation / Prototype



Verdict screen: type, confidence, severity, and what to do next

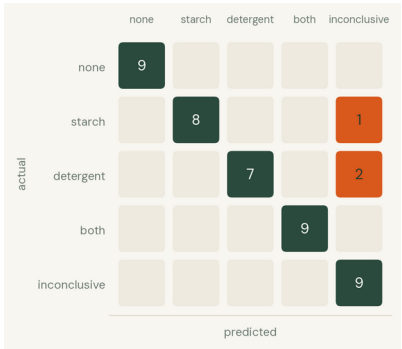


A clean sample comes back as none, not as a blank screen



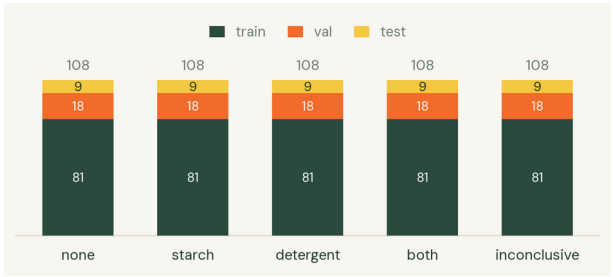
Evidence log: every test stamped, each with a PDF and QR report

HELD OUT BY LIGHTING, NOT BY IMAGE



45 samples, 3 lighting conditions the model never saw in training

TRAINING SET, 534 SYNTHETIC STRIP IMAGES



Splits are assigned by lighting condition, so no condition appears in two splits. The model returns inconclusive on 12 of the 45 held-out samples: 9 are the 9 labelled inconclusive, and the other 3 are one starch and two detergent it declined to call rather than guess. On the 33 it does answer, it is right every time.

6.0 ms
median analysis time, in-process (n=20)

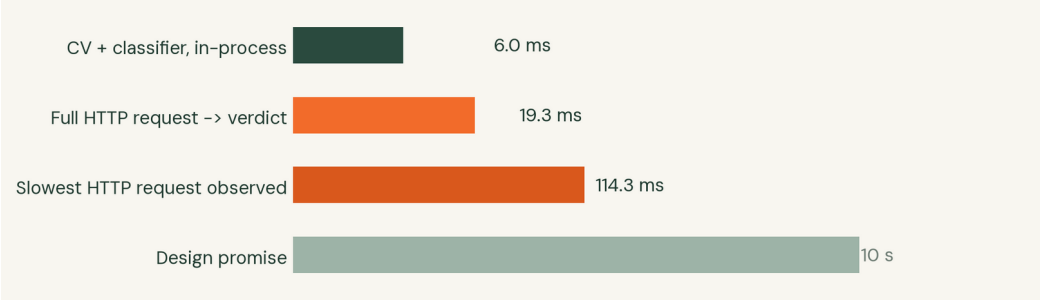
19.3 ms
full HTTP request to verdict (n=12)

0.42 s
cold start, API ready to serve

291 tests · 90% coverage
passing, ruff clean

Feasibility & Impact

MEASURED SPEED, AGAINST THE 10-SECOND PROMISE



WHAT THE MODEL DOES WITH AMBIGUITY



Held-out by lighting condition · n = 45 · selective accuracy 1.000 · coverage 0.733

Both bars are the same run: 33 answered, 12 declined.

Cost per test (estimate)

Printed card, two zones	**≈ ₹2**
One iodine or MB drop	**≈ ₹1**
Phone and network	**already owned**

≈ ₹3 per test, excluding the phone

Estimate only. Reagent prices are pending; the kit is being bought this week and the figure will be restated once.

What a vendor actually needs

Printed cards, two dropper bottles and a phone. No app store, no account, no internet after the first page load, no calibration by hand.

The same card and app work for milk, water and any liquid the two reagents screen for.

Impact, and the road to 24 October

Every test carries a batch id and a QR, so a distributor can scope a recall to the batches that failed instead of pulling a whole product line.

23 Oct real strips captured and the model retrained · 24 Oct final, live on real reagents.

Research & References

Research papers & patents

Fragkos N., Bouzembrak Y., Erasmus S. W. (2026)

The role of artificial intelligence in combating food fraud: a systematic literature review. Critical Reviews in Food Science and Nutrition. doi:10.1080/10408398.2026.2654710

Food Chemistry 468:142439 (2025)

Recent advances on AI-based approaches for food adulteration and fraud detection: challenges and opportunities. PMID 39675268

Data, reports & regulations

FSSAI, tabled in Parliament 24 July 2026

Lok Sabha Unstarred Question No. 1092. 14,668 of 39,351 milk and dairy samples non-conforming in FY 2025-26.

WHO, Food safety fact sheet, 4 June 2026

866 million illnesses, 1.52 million deaths and US$ 310 billion a year; children under 5 carry 29% of the burden.

Tools, datasets & field work

Open-source stack

OpenCV, scikit-learn, ONNX Runtime, FastAPI, SQLite, Python 3.12. Model trained on an RTX 4050, served on CPU.

Dataset and field work

534 synthetic strip images across 5 lighting conditions, plus printed colour targets photographed to validate the imaging half before any chemistry. Real iodine and methylene-blue strips captured 23 October, the night before the final.

How the sources were used

WHO and FSSAI figures set the size of the problem on slide 2 and are quoted with their publication dates. The two review papers informed the decision to keep a rule-table fallback beside the learned model, since both reviews flag model opacity as the main barrier to regulatory adoption. No figure in this deck was rounded into existence; every number traces to a file in the repository or to a cited publication.

Conclusion

A printed card and a phone camera, turned into a repeatable adulteration screen that keeps its own record.

291
tests passing, 90% coverage

6.0 ms
median analysis time

5
verdict classes, one of them unsure

0
fabricated verdicts by design

What is built and running

Capture, calibration, zone measurement, classification, abstention, evidence log, PDF and QR export, and a browser UI that works on a phone. The repository holds 34 commits of test-driven work and starts from a clean checkout in one command.

What we are honest about

Colour thresholds are validated on synthetic reagent scenes, not yet on real chemistry. The 24 October final runs on real iodine and methylene-blue strips and the model is retrained on those photos first. A reading is a screening signal, not a legal test result.

ROADMAP TO 24 OCTOBER

14 Oct
Shortlist announced

15-22 Oct
Printable targets, mid-grey backdrop and frame guide built

23 Oct
Real strips captured, model retrained, full rehearsal

24 Oct
Final, 09:00-12:00 at PCCOE, live on real reagents

Thank You

[Team Name] · [Contact email]